# Performance Audit Tools

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Img src: https://bit.ly/ng-tools-img

# Performance Audit Tools

- PageSpeed & Lighthouse
- WebPageTest.org

- Source Map Explorer
- Webpack Bundle Analyzer
  - Vite Bundle Visualizer
- Import Graph Visualizer
- Chrome DevTools
- Angular DevTools Profiler

# Performance Audit Tools

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Initial Load

Bundle Size

Runtime

**Focus**

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# Web Audit Tools

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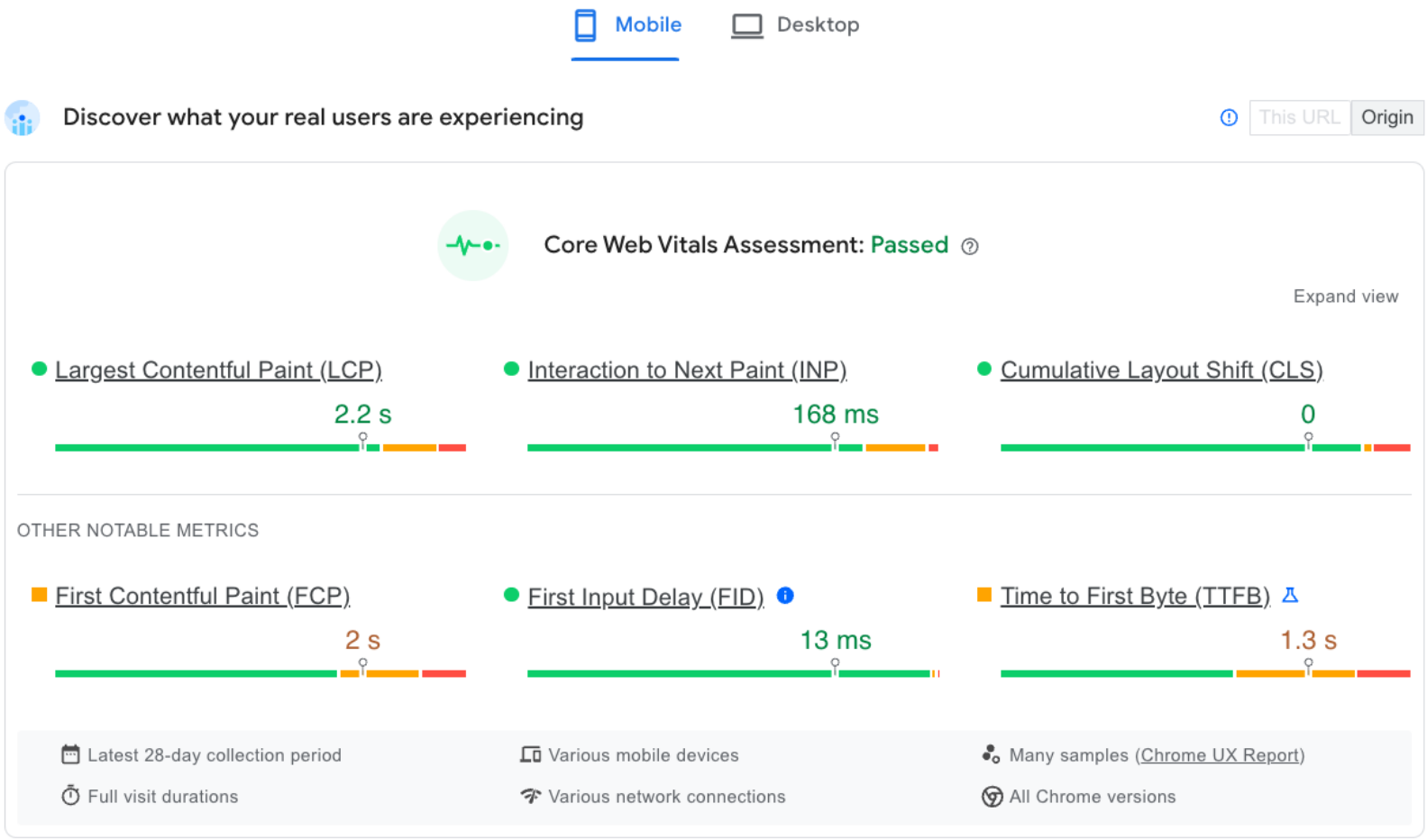
# PageSpeed Insights vs Chrome Lighthouse

- PageSpeed Insights
- **Real** user data & **Lighthouse** lab results
- Go to pagespeed.web.dev & enter URL
- Test is being run on Google servers
- Performance, new since 2023
  - Accessibility
  - Best Practices
  - SEO

- Chrome Lighthouse
- Open URL in your browser
  (run in incognito mode, close other Apps)
- Open DevTools Lighthouse tab → Analyze
- Performance, other tests
  - Accessibility
  - Best Practices
  - SEO
  - **PWA**

# PageSpeed Score – Real Users (Origin)



# PageSpeed Score – Lab Data (URL)

Diagnose performance issues

| Performance | Accessibility | Best Practices | SEO |
|---|---|---|---|
| 90 | 87 | 100 | 86 |

90

Performance

Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.

▲ 0–49 ■ 50–89 ● 90–100

METRICS

Expand view

| | |
|---|---|
| ● First Contentful Paint 1.7 s | ■ Largest Contentful Paint 3.0 s |
| ■ Total Blocking Time 200 ms | ● Cumulative Layout Shift 0.007 |
| ● Speed Index 2.9 s | |

Captured at Apr 22, 2024, 8:22 PM GMT+2
Initial page load
Emulated Moto G Power with Lighthouse 11.5.0
Slow 4G throttling
Single page session
Using HeadlessChromium 122.0.6261.94 with lr

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# PageSpeed Score – Case Studies

Good examples

- Websites
  - WP Rocket
  - Wikipedia
- Angular Apps
  - https://www.emilfrey.ch/

# Demo

## PageSpeed Insights

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# Chrome Lighthouse – Getting Started

- Lighthouse is tool to test NG App **during development**
- We still need a **production build** to get useful results
- How can we run a production build on localhost?

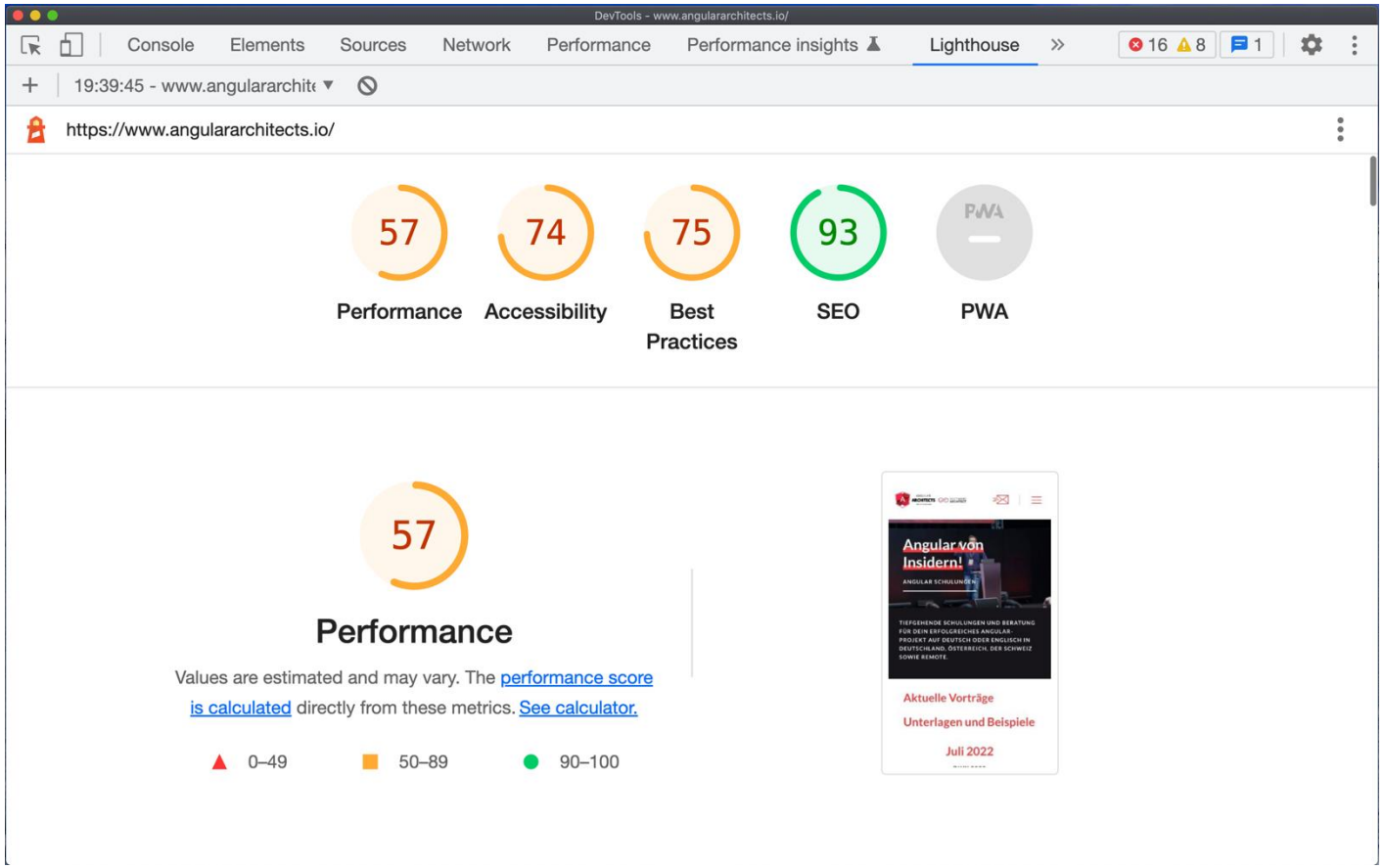
# Chrome Lighthouse – Prod Build Localhost

1. Prod build → ng b(uild)

2. Serve the build on your localhost with
   - Using localhost tool like MAMP / WAMP / XAMPP
   - NPM serve
     - https://www.npmjs.com/package/serve
   - Or edit hosts file manually if you know how to do that ☺

3. Open in Chrome and run Lighthouse

# Chrome Lighthouse – Summary



# Chrome Lighthouse – Details

METRICS

Expand view

| | |
|---|---|
| First Contentful Paint<br>2.1 s | Time to Interactive<br>7.7 s |
| Speed Index<br>3.8 s | Total Blocking Time<br>230 ms |
| Largest Contentful Paint<br>6.5 s | Cumulative Layout Shift<br>0.054 |

View Original Trace

View Treemap

Show audits relevant to: All FCP TBT LCP CLS

# Demo

## Chrome Lighthouse

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# Important Metrics

What's being measured?

- Time to First Byte (TTFB)
- **First Contentful Paint (FCP)**
- **Speed Index** (originally by WebPageTest)
- **Largest Contentful Paint (LCP)**
- Time to Interactive (TTI)
- **Total Blocking Time (TBT)** → TTI - FCP
- **Cumulative Layout Shift (CLS)**
- First Input Delay (FID)
- Interaction to Next Paint (INP) → new!

# WebPageTest.org

- An alternative to Google's PageSpeed & Lighthouse
- Measures same things (TTFB, FCP, Speed Index, LCP, TBT, CLS)
- Generates waterfall, screenshot, video & a content breakdown
- Runs multiple tests at once (e.g. 3 or 5)
- Choose test location and configure test machine
- Lots of further advanced settings, like HTTP Authentication

# WebPageTest – Summary

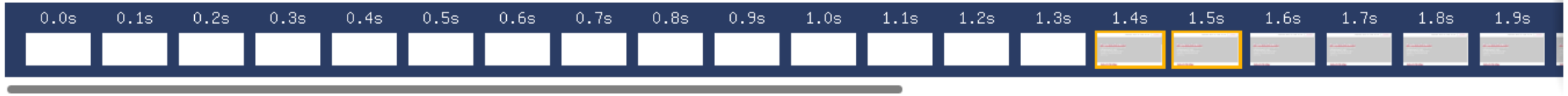
## Observed Metrics (Based on Median Run by: ▸ Speed Index)

Note: Metrics offered will vary

FIRST VIEW (RUN 1)

| First Byte | Start Render | FCP | Speed Index | LCP | CLS | TBT | Total Bytes |
|---|---|---|---|---|---|---|---|
| .479 s | 1.400 s | 1.370 s | 3.112 s | 3.307 s | .002 | .482 s | 1,976 KB |

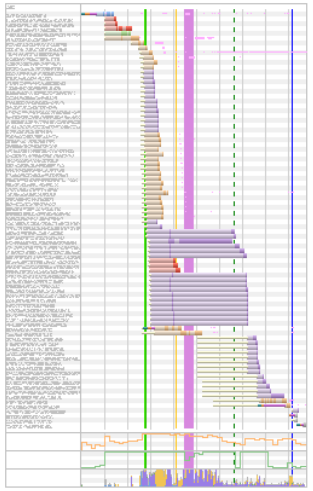
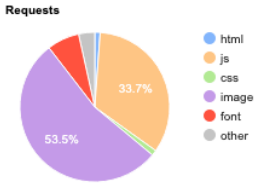
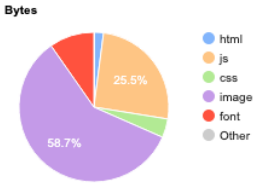
Visual Page Loading Process (Explore)




Compare First Views

Plot Full Results

# WebPageTest – Details

| | Waterfall | Screenshot | Video |
|---|---|---|---|
| First View<br>(4.583s)<br><br>Timeline (view)<br>Processing Breakdown<br><br>Trace (view) |  | | Filmstrip View<br>-<br>Watch Video |
| Content Breakdown |   |   | |

# Demo

## WebPageTest

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# Lab 01 Web Audit Tools

PageSpeed / WebPageTest / Lighthouse

# Build Analysis Tools

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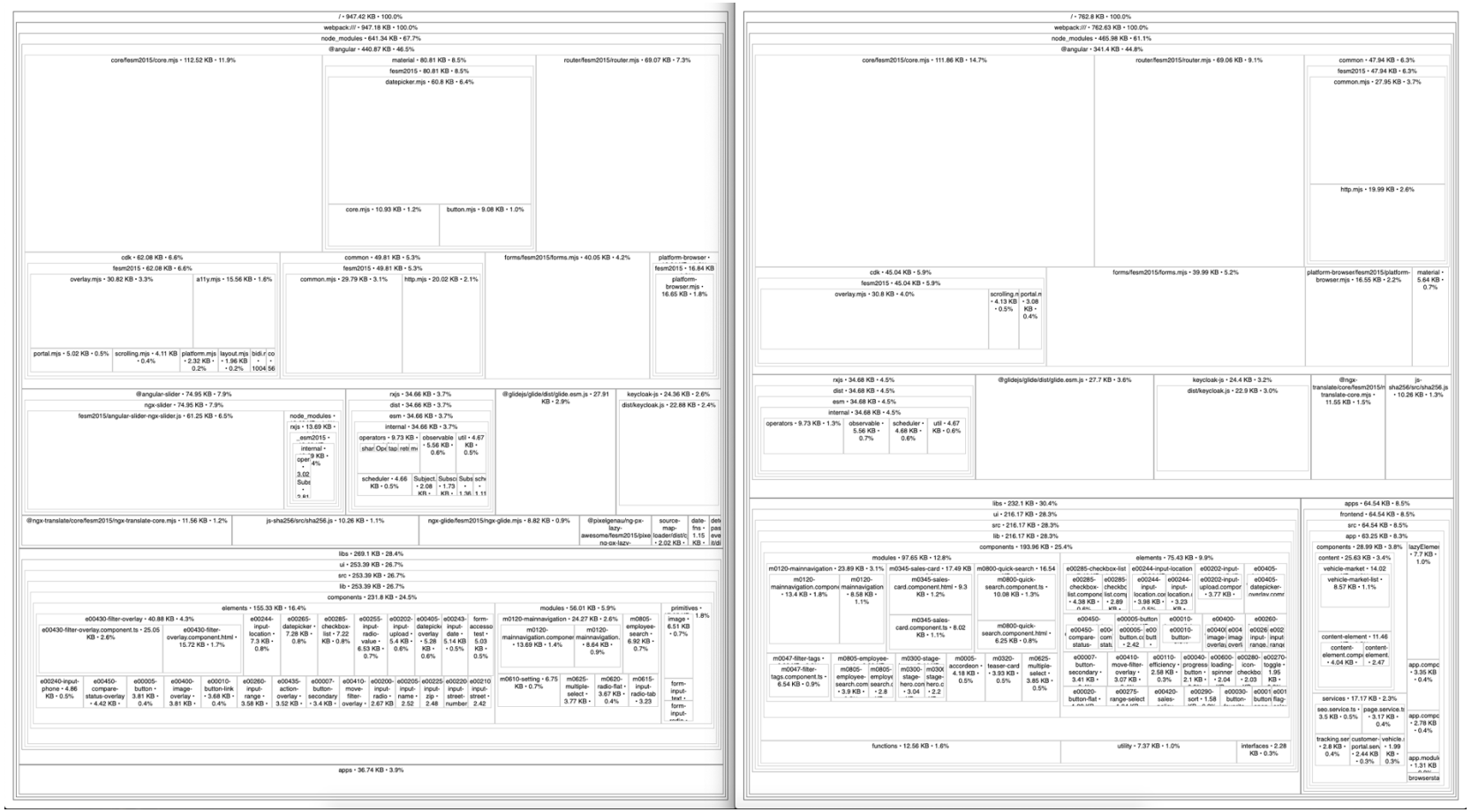
# Source Map Explorer

- Needs generated source maps to work
  - Either set in build options (angular.json)
  - Or just use the build flag "--source-map"
- Analyzes a single js file or whole bundle / build
  - main bundle
  - or lazy loading bundles
- Determines where each byte in your code comes from
- Shows a black/white treemap visualization of build size

# Source Map Explorer



# Demo

## Source Map Explorer

https://www.npmjs.com/package/source-map-explorer

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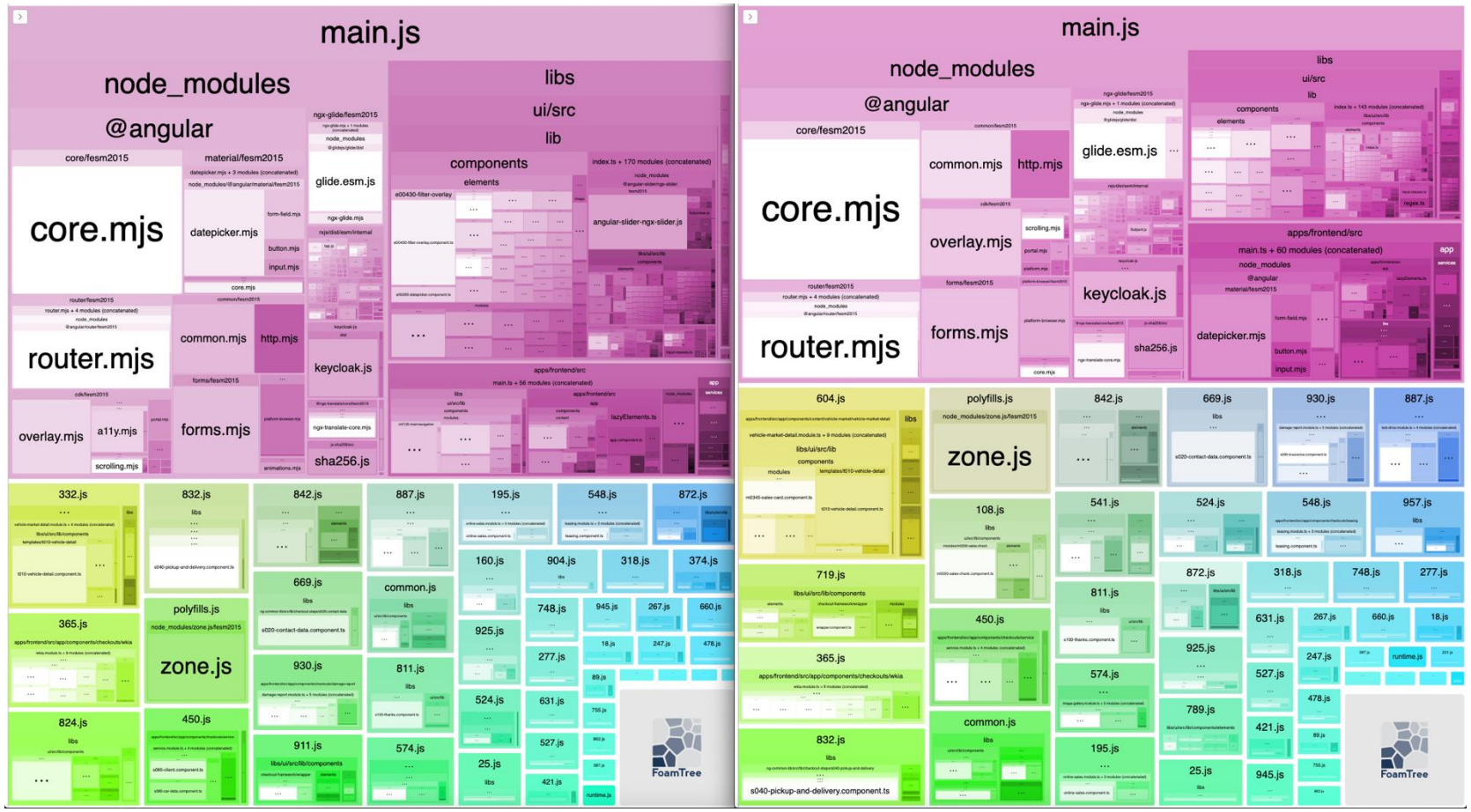
# Webpack Bundle Analyzer

- Needs a generated stats.json to work
  - Either set in build options (angular.json)
  - Or just use the build flag "--stats-json"
- Analyzes the whole build
- Visualize size of all js chunks
  - Good to analyze lazy loading
- Interactive, zoomable and colorful treemap ☺

# Webpack Bundle Analyzer



# Demo

## Webpack Bundle Analyzer

https://github.com/webpack-contrib/webpack-bundle-analyzer

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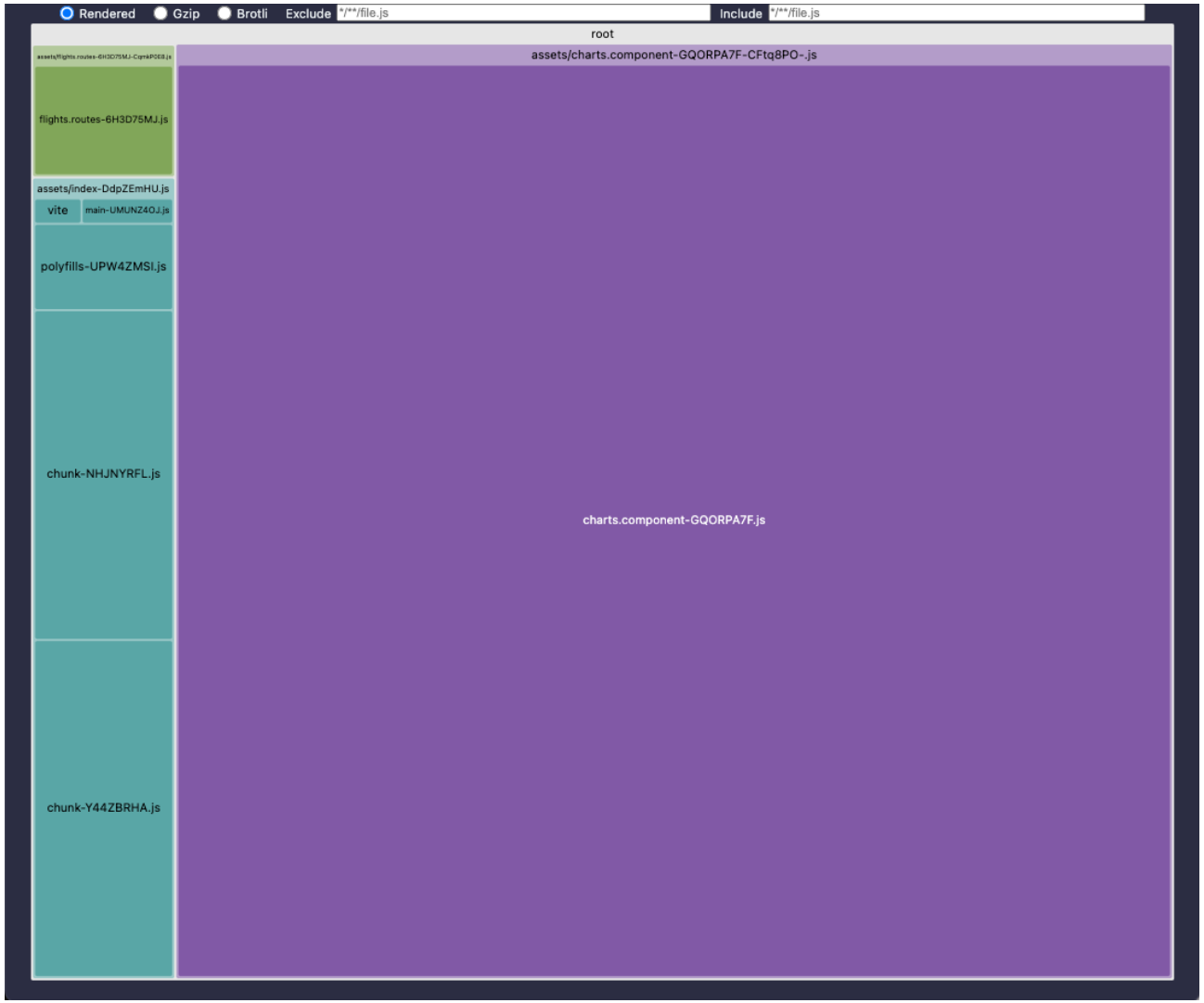
# Vite Bundle Visualizer

- Similar to Webpack Bundle Analyzer but for vite
- Also analyzes the whole build
- Also visualizes size of all js chunks
  - Also good to analyze lazy loading
- Interactive, zoomable but not so beautiful treemap ☺

# Vite Bundle Visualizer



# Demo

## Vite Bundle Visualizer

https://www.npmjs.com/package/vite-bundle-visualizer

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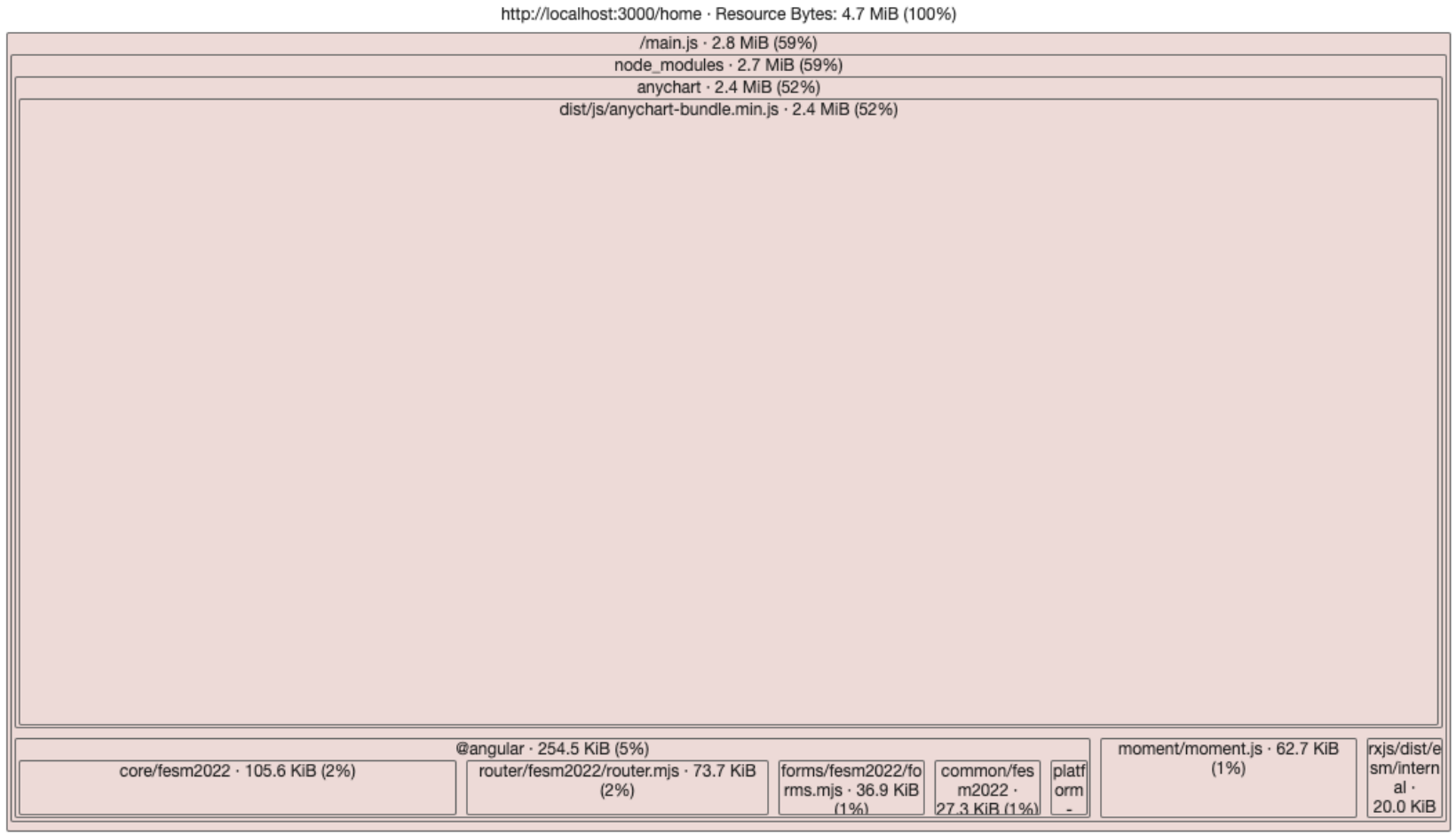
# Lighthouse Treemap

- Also similar to Webpack Bundle Analyzer
- Also analyzes the whole build
- But generated by Chrome – no extra build needed ☺

# Lighthouse Treemap



# Demo

## Lighthouse Treemap

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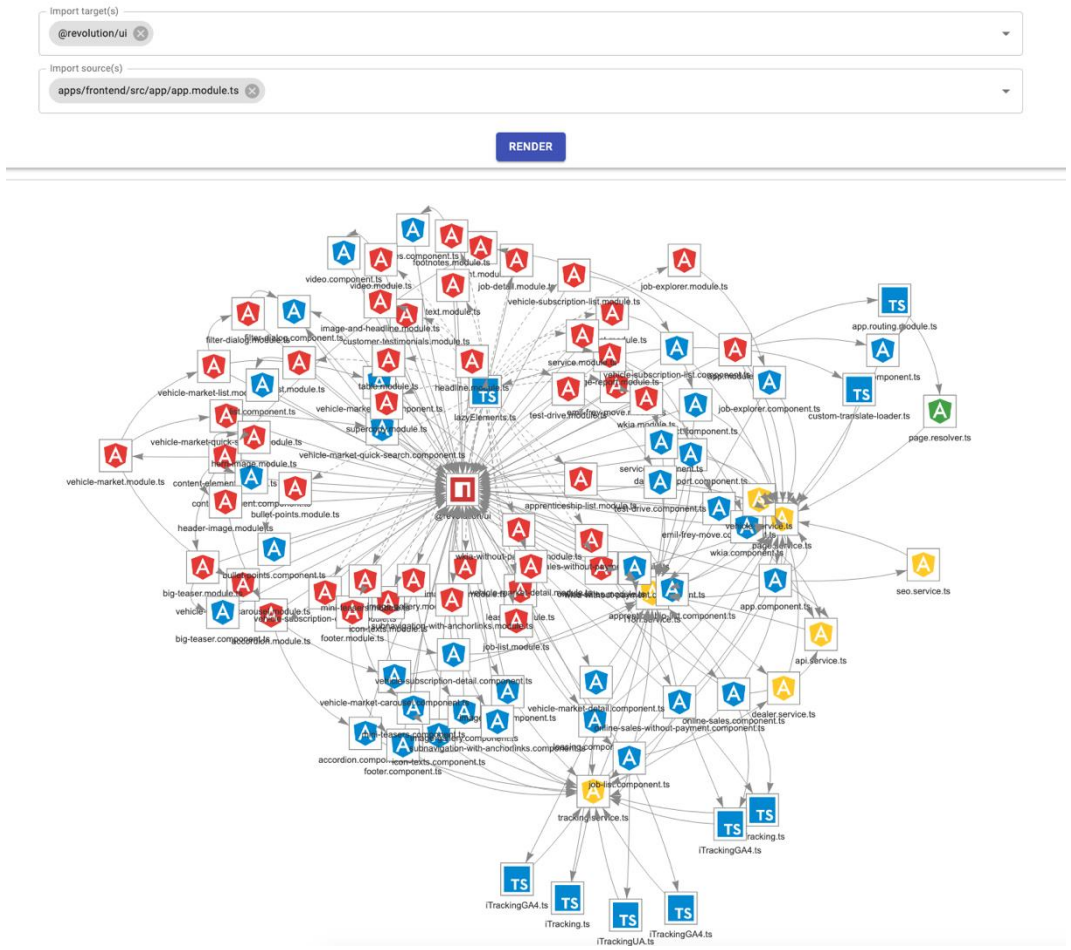
# Import Graph Visualizer

– A development tool for filtering and visualizing import paths within a JavaScript/TypeScript application

– Allows filtering import paths by source & target modules

– Allows zooming in to a limited subsection of your app, which will likely be easier to analyze than the entire app

# Import Graph Visualizer



# Import Graph Visualizer – How To

- npx import-graph-visualizer --entry-points path/to/entry/module --ts-config path/to/tsconfig

- e.g. npx import-graph-visualizer --entry-points src/main.ts --ts-config tsconfig.app.json

# Demo

## Import Graph Visualizer

https://github.com/rx-angular/import-graph-visualizer

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# Lab 02 Build Analysis

Source Map Explorer / Webpack Bundle Analyzer / Vite Bundle Visualizer

# Runtime Audit Tools

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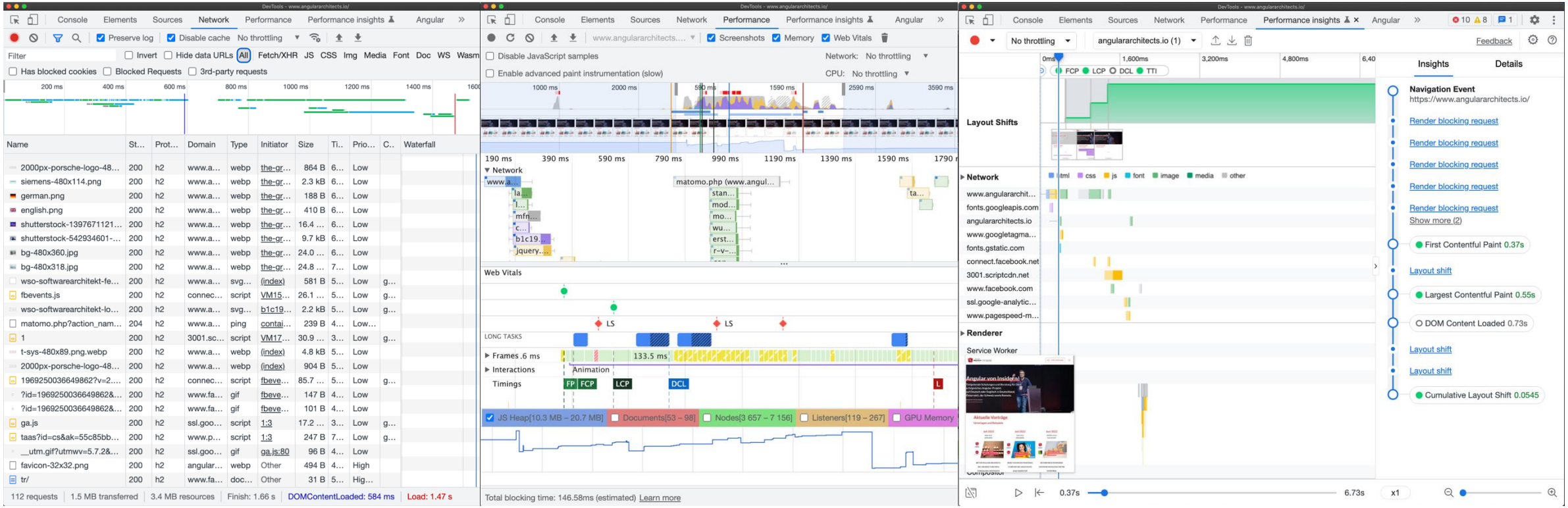
# Google Chrome DevTools

- The Chrome DevTools are not only used for
  - Styling (Elements)
  - Debugging (Console)

- But also for Performance
  - Network
  - Performance
  - Memory
    - memory heap comparison
  - Performance Insights (Beta!)

# Google Chrome DevTools



# Demo

## Chrome DevTools

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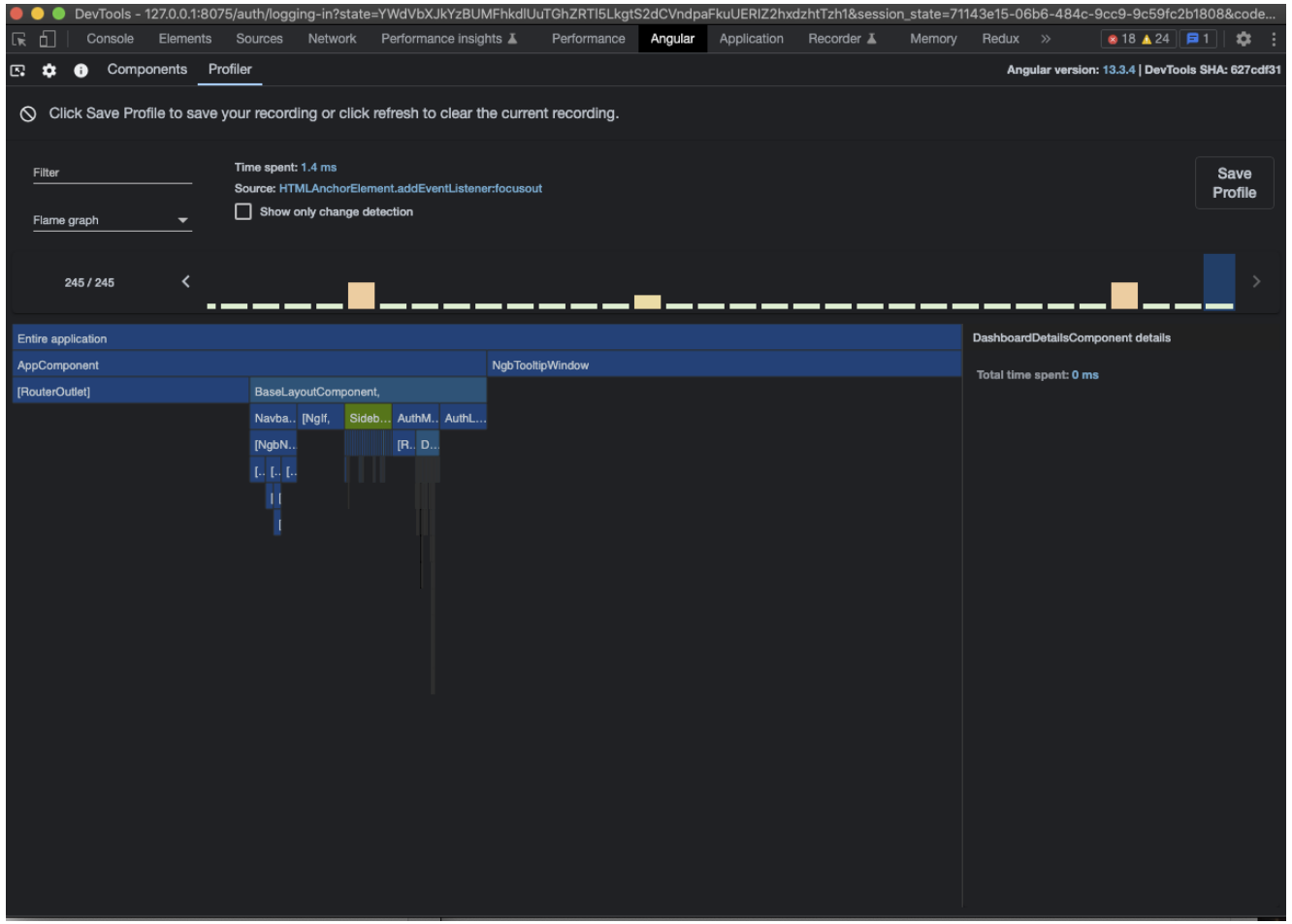
# Angular DevTools Profiler

- Angular DevTools extension can be added to Chrome
  - https://chromewebstore.google.com/detail/angular-devtools/ienfalfjdbdpebioblfackkekamfmbnh
  - Features a **Component Tree** to inspect the components
  - **Profiler**
  - And the newly added **Injector Tree** (Dependency Injection)

- Profiler shows individual change detection (CD) cycles
  - What triggered CD
  - How much time it took executing CD

# Angular DevTools Profiler



# Demo

## Angular DevTools Profiler

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Initial Load

Bundle Size

Runtime

Recap

# References

- Google Web Dev
  - https://pagespeed.web.dev
  - https://web.dev/metrics/
- Improving Load Performance - Chrome DevTools 101
  - https://www.youtube.com/watch?v=5fLW5Q5ODiE
- How to analyze your JavaScript bundles
  - https://www.youtube.com/watch?v=MxBCPc7bQvM

# Questions?

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